

Modeling and Control of a PEM Fuel Cell Stack

From physical principles to per-cell calibration and PI-based power control

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Outline

- 1 Motivation & Objectives
- 2 Experimental Platform
- 3 Background
- 4 Simplified Model
- 5 Identification & Analysis
- 6 Experiments & Results
- 7 Control Architecture
- 8 Conclusion

One-line storyline

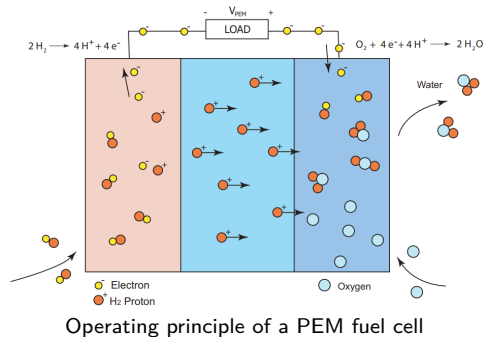
From an electrochemical model → per-cell calibration → dynamic characterization → PI-based power control.

Motivation — Why Control a PEM Fuel Cell?

- $\text{H}_2 \rightarrow$ electricity, only by-products **water + heat** — clean transport / stationary power.
- PEMFC wins on **low temperature, high power density, fast start-up**.
- Unmanaged operation fails:
 - reactant starvation $\rightarrow$ irreversible degradation;
 - excess anode-cathode pressure $\rightarrow$ membrane damage.
- Cells differ within a stack (deviations up to $\approx 7\%$) — a stack is *not* one ideal cell.

$\Rightarrow$ Feedback control is required — and it needs a model that is simple to analyze yet faithful to the hardware.

Ref.: [3] Daud et al. 2017; [1] Andrea et al. 2006

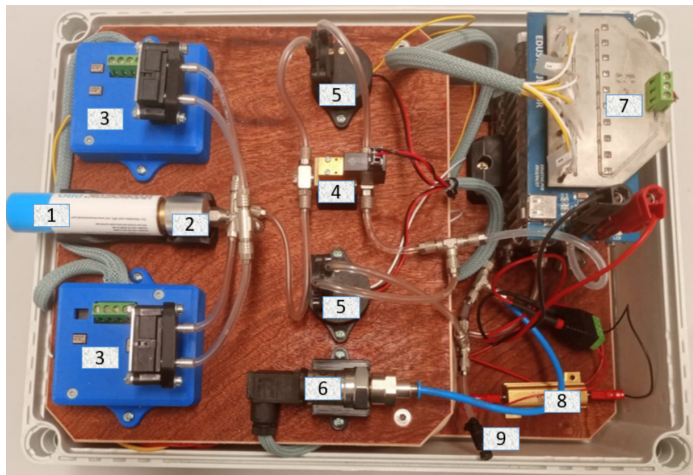


- ① Simplified **control-oriented dynamic model** of a cell and the 10-cell stack.
- ② **Per-cell parameter identification** from bench data.
- ③ **Linearized state-space** model + controllability / observability analysis.
- ④ **Experimental** dynamic response to resistive load steps (timescales for control bandwidth).
- ⑤ A **PI-based control architecture** through a DC/DC boost converter.

Thread of the talk

Each result below maps back to one of these five objectives.

Experimental Setup — The Bench



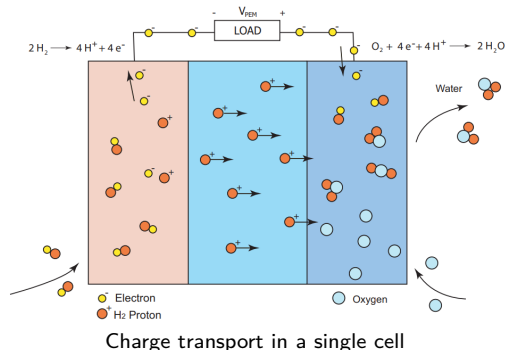
- ① H₂ storage (metal-hydride)
- ② Pressure regulator
- ③ Flow sensor (Honeywell AWM42)
- ④ Proportional solenoid valve
- ⑤ Pressure sensor (MPX4250DP)
- ⑥ Relative pressure sensor
- ⑦ **10-cell EDUSTAK JUNIOR stack**
- ⑧ Load resistors (25/3.3/6.8 Ω)
- ⑨ Mounting / DAQ base

Instrumented PEMFC bench (numeric call-outs 1–9)

10-cell stack, per-cell voltage acquisition at 1 Hz via a serial DAQ chain built in the project.

Half-reactions

- Anode: $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
- Cathode: $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$
- Overall: $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{electricity} + \text{heat}$
- Protons cross the membrane; electrons drive the load.
- MEA = elementary repeating unit of the stack.
- **Water management is critical:** dry $\rightarrow$ low conductivity, flooded $\rightarrow$ blocked channels.



Ref.: [9] Pukrushpan 2003

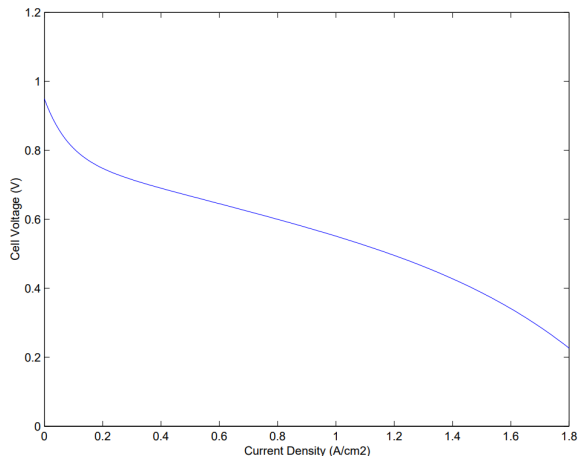
Voltage Losses & Polarization Curve

Three loss regimes shape the curve:

- **Activation** — low current (reaction kinetics)
- **Ohmic** — mid range ($\approx$ linear)
- **Concentration** — high current (mass transport)

The polarization curve = cell voltage vs. current density = the backbone of the model.

Ref.: [1] Andrea et al. 2006



Polarization curve (losses bend the ideal flat voltage)

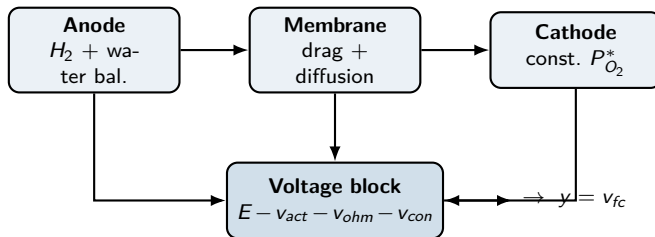
- Natural actuator = **hydrogen supply**, but usable pressure is **narrowly bounded** (regulator cap + membrane-damage risk).
- The valve **saturates** before meeting the load demand → too little authority for direct current control.

Decision

Move control to the electrical side: stack → **DC/DC boost converter** → fast **PI loop**.

Roles: *electrical control for the fast dynamics, gas supply for the slow operating conditions.*

Ref.: [16] Zhu et al. 2017; [6] Kharrat & Haußmann 2025



States / inputs / output

- $x = (P_{H_2}, m_{w,an}, m_{w,ca})$
- $u = (i_{FC} [\text{dist.}], W_{H_2,in} [\text{man.}]), y = v_{fc}$

Assumptions

- H_2 + air only; ideal gas; slow T
- $P_{O_2}^* = 0.21 P_{atm}$ fixed

Ref.: [9] Pukrushpan 2003; [11] Thanapalan et al. 2008

Open-circuit (Nernst) potential:

$$E = 1.229 - 0.85 \times 10^{-3} (T_{fc} - 298.15) + 4.31 \times 10^{-5} T_{fc} \left(\ln p_{H_2} + \frac{1}{2} \ln p_{O_2} \right)$$

Cell voltage = reversible voltage – three semi-empirical losses:

$$v_{fc} = E - \underbrace{v_{act}}_{\text{activation}} - \underbrace{v_{ohm}}_{\text{ohmic}} - \underbrace{v_{con}}_{\text{concentration}}$$

Calibrated unknowns

Six parameters per cell $\theta = (\xi_1, \xi_2, \xi_3, \xi_4, B, c)$; stack voltage $V_s = \sum_{\text{cells}} v_{fc}$.

Ref.: [11] Thanapalan et al. 2008; [2] Cao et al. 2019 — full derivation in backup

Anode (hydrogen pressure):

$$\dot{P}_{H_2} = \frac{R_{H_2} T}{V_{an}} \left(W_{H_2, in} - W_{H_2, out} - \frac{M_{H_2} n}{2F} i_{FC} \right)$$

- **Membrane:** electro-osmotic drag + back-diffusion; Springer sorption isotherm $\lambda(a)$.
- **Cathode:** constant $P_{O_2}^*$; water balance with reaction-generated water.

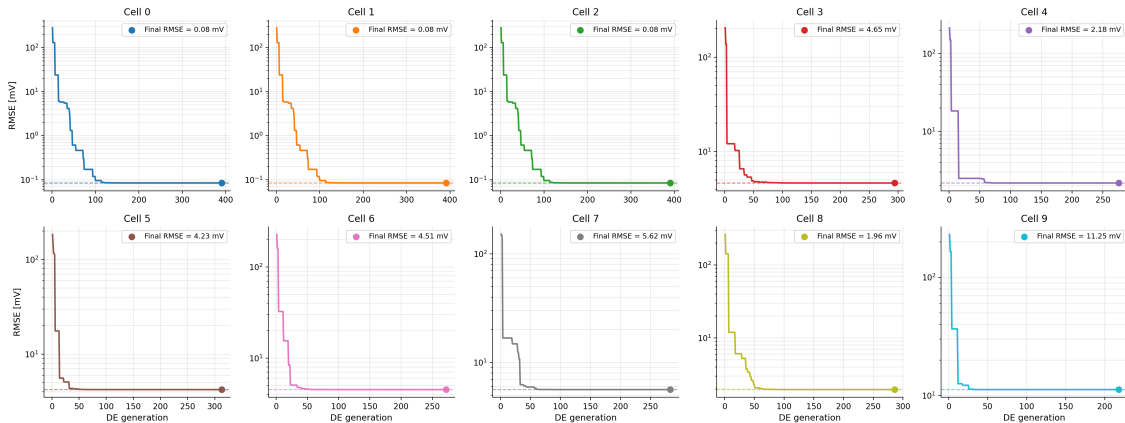
Simplifications (cost of being control-oriented)

No spatial gradients, no oxygen transient, no liquid water $\rightarrow$ low order, fast to simulate and linearize.

Ref.: [14] Weber & Newman 2004; [15] Zhang et al. 2017

Per-Cell Parameter Identification — DE Convergence

Differential-evolution convergence per cell — Optimization Experiment



RMSE vs. generation (log scale) for the 10 cells; dashed line = final RMSE after L-BFGS-B refinement. Most cells converge in a few tens of generations.

Goal: fit the six parameters $\theta = (\xi_1, \xi_2, \xi_3, \xi_4, B, c)$ of each cell.

- Minimize **RMSE** between model and measured cell voltage over a calibration window.
- Two-stage optimizer: *differential evolution* (global; pop. 20, ≤ 800 gen., fixed seed) $\rightarrow$ *L-BFGS-B* (local refinement).
- Bounds encode physics: $\xi_4 \leq 0$, $B \geq 0$, $c \geq 0$.

Final per-cell RMSE: 2.0–11.3 mV for the healthy cells (mean 8.0 mV).

Ref.: [2] Cao et al. 2019; [11] Thanapalan et al. 2008

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}, \quad y = C\mathbf{x} + D\mathbf{u}, \quad \mathbf{x} = \begin{bmatrix} P_{H_2} \\ m_{w,an} \\ m_{w,ca} \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} i_{FC} \\ W_{H_2,in} \end{bmatrix}$$

A, B — read off directly

Hydrogen-pressure + two water balances are *linear* at constant T .

C, D — Jacobian linearization

Voltage output is *nonlinear* (logs of P_{H_2}, i_{FC} ; membrane resistivity) $\Rightarrow$ first-order Taylor expansion.

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\beta & \gamma \\ 0 & \beta & -\gamma \end{bmatrix}, \quad B = \begin{bmatrix} -\frac{R_{H_2} T M_{H_2} n}{2 F V_{an}} & \frac{R_{H_2} T}{V_{an}} \\ d_1 & 0 \\ d_2 & 0 \end{bmatrix}$$

Controllability

Controllable for all physically realistic electro-osmotic drag coefficients n_d (singular only at one degenerate condition $n_d = \frac{nV_{an}}{2(V_{an}+V_{ca})}$).

Observability (voltage-only output)

Voltage **observes P_{H_2} but not the water states** $\Rightarrow$ a **reduced-order voltage-based observer** suffices for hydrogen-pressure regulation; full-state would need extra sensors.

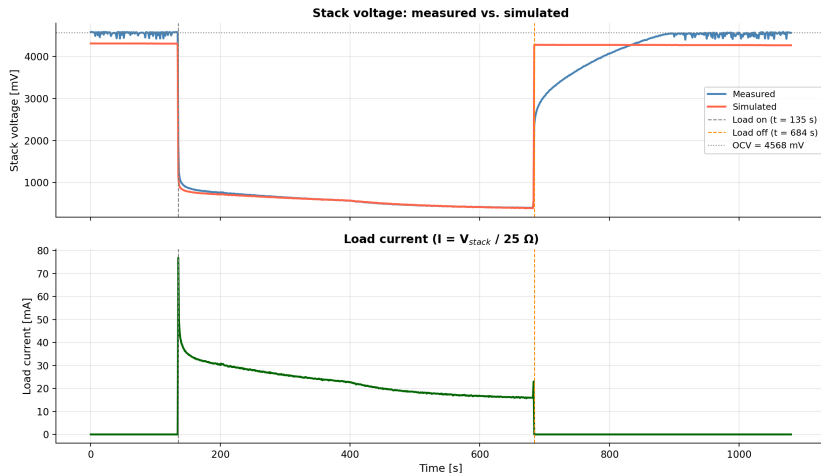
Consequence: a voltage measurement is enough to track the pressure state we actually want to regulate.

- 10-cell stack; per-cell voltage at **1 Hz** via the project's serial DAQ chain.
- **Optimization Experiment** (Apr 2026): $25\ \Omega$ connected at $t = 135\text{ s}$, removed at $t = 684\text{ s} \rightarrow$ calibration data.
- **TR-current steps** (9 Jun 2026): $3.3\ \Omega$ and $6.8\ \Omega$ from open circuit; fit $I(t) = I_{ss} + \Delta I e^{-t/\tau_I}$.

Two experiments, two purposes

Optimization Experiment $\rightarrow$ *static* calibration; TR-current steps $\rightarrow$ *dynamic* timescales for control.

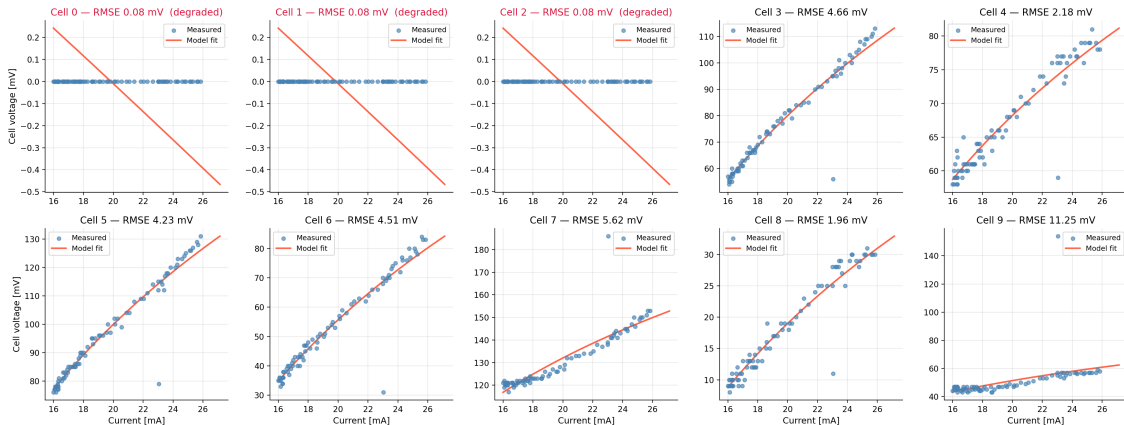
Result — Stack Voltage & Load Current



- OCV $\approx 4.57\text{ V} \rightarrow \approx 0.8\text{ V}$ under the $25\ \Omega$ load; simulation **tracks** the measurement across the loaded window.
- Per-cell fits aggregate correctly to the stack level.

Result — Per-Cell V-I Calibration Fits

Optimization Experiment — per-cell voltage-current calibration fits

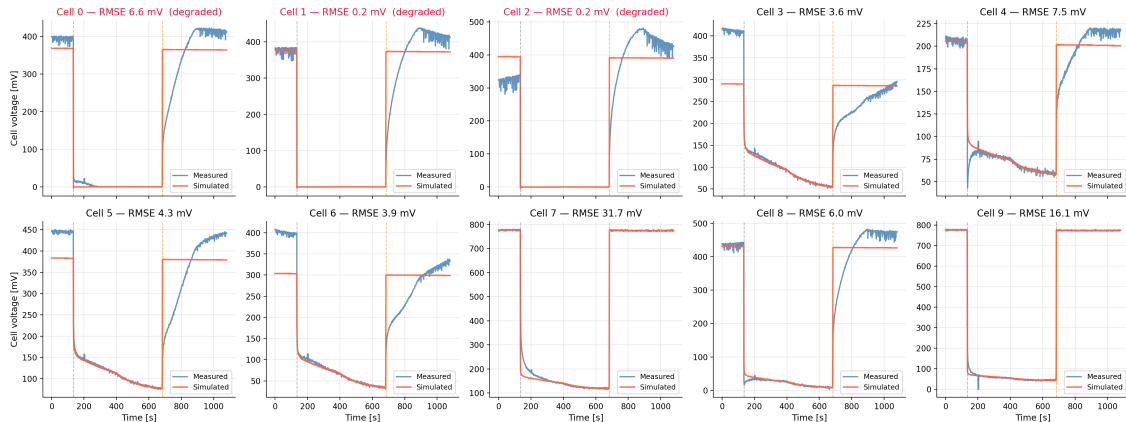


Measured points (blue) vs. identified model (red), all 10 cells. Per-cell calibration RMSE in each panel title.

- Healthy cells matched with calibration RMSE **2.0–11.3 mV** (all below 12 mV).
 - Concentration (B) and contact-resistance (c) terms stay ≈ 0 .
- ⇒ **Activation and ohmic effects dominate** the covered operating range.
- Six parameters per cell are enough to reproduce each healthy cell.

Result — Per-Cell Voltage: Measured vs. Simulated

Optimization Experiment — measured vs. simulated per-cell voltage (load on at $t = 135$ s, off at $t = 684$ s)



Measured (blue) vs. simulated (red) over the full experiment, all 10 cells. Per-cell time-series RMSE in each panel title.

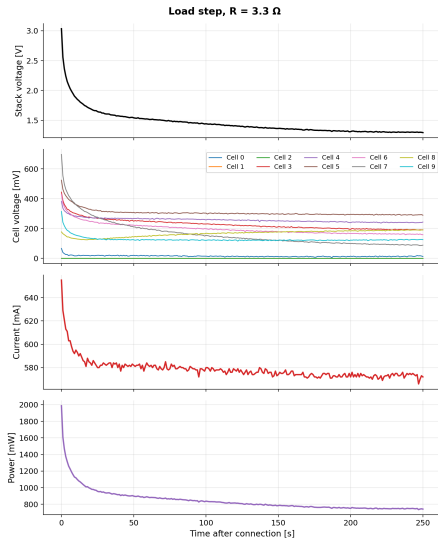
- Healthy cells (3–9) tracked closely across the loaded window.
- **Mean time-series RMSE = 8.0 mV** for the healthy cells.
- Cells 0–2 stay near zero under load → degraded (next slide).
- Larger residuals on Cells 7 and 9 come from the load-on transient, excluded from the calibration window.

Result — Identification Doubles as Diagnosis

- Cells 0, 1, 2 collapse to $\approx 0\text{ V}$ under load (top-left panels of the previous figure).
- The optimizer returns **identical, degenerate** parameter sets for them (RMSE $< 0.1\text{ mV}$ against a flat signal).
- This is a **dead-electrode signature**, not a physically meaningful fit — no parameters within the bounds can reproduce a dead cell.

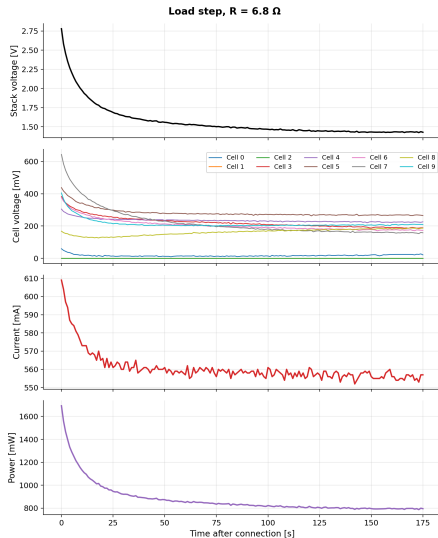
Takeaway

The calibration is also a **health check**: it flags faulty cells automatically. Full per-cell parameter table in **backup B2**.



Four channels after the step: stack voltage, individual cell voltages, current, and power.

Monotonic relaxation, no overshoot $\Rightarrow$ first-order behavior.

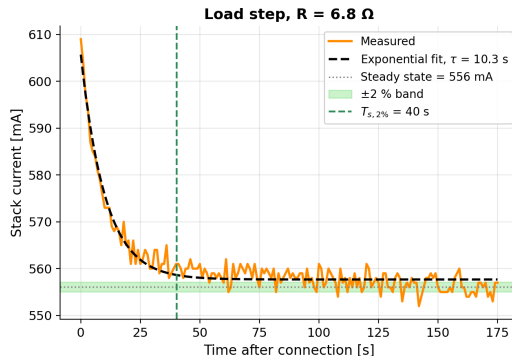
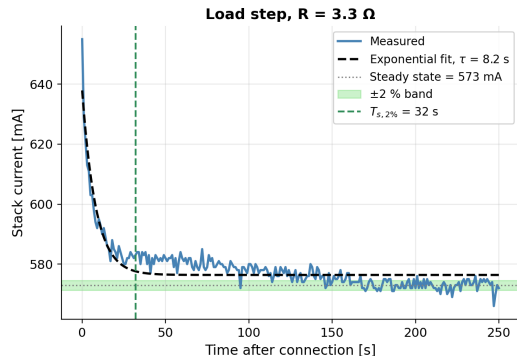


Same four channels under the lighter load.
The current settles more slowly than at 3.3Ω .

Same first-order shape, longer time constant.

Result — Current Step Response & Time Constants

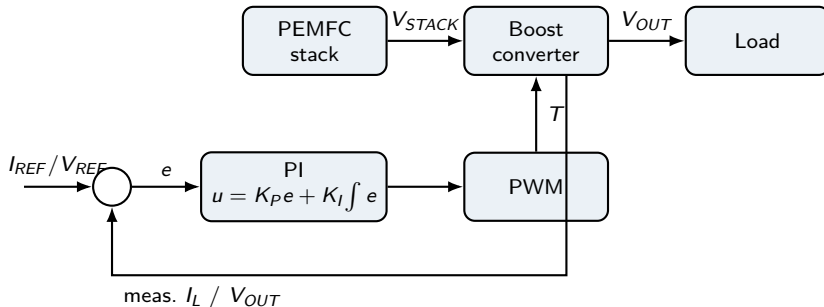
Current step response to a resistive load



	3.3Ω	6.8Ω
τ_I [s]	8.2	10.3
ΔI [mA]	82.1	52.9
$T_{s,2\%}$ [s]	32.0	40.2

- Lighter load $\Rightarrow$ slower (lower current density).
- **Control target: act well within $\tau_I \approx 8\text{--}10$ s.**

DC/DC Boost Converter + PI Controller



Key idea

The stack is *neither* an ideal current nor voltage source — the **regulated converter output** is shaped into one by the PI loop on the duty cycle T .

Ref.: [4] Derbeli et al. 2017; [16] Zhu et al. 2017

Current mode (protect the stack)

- Regulate $I_L \rightarrow I_{REF}$; $e = I_{REF} - I_L$.
- $I_{REF} = 9.74 \text{ A} = \text{max-power point (50 W stack; 10 A hard limit)}$.
- Prevents **overcurrent**.

Voltage mode (fixed load voltage)

- Regulate $V_{OUT} \rightarrow V_{REF}$; $e = V_{REF} - V_{OUT}$.
- Same averaged converter model.
- Natural for **output-voltage** supply.

Same PI structure & converter model — differ only in the feedback variable. Online (Ziegler–Nichols-style) tuning, no plant transfer function needed.

Ref.: [4] Derbeli et al. 2017

- Six-parameter model fitting all healthy cells $< 12 \text{ mV}$; built-in **diagnosis** of degraded cells (0,1,2).
- Explicit **state-space** model: controllable; P_{H_2} observable from voltage alone.
- Experimental dynamics: $\tau_I \approx 8\text{--}10 \text{ s}$, $T_{s,2\%} \approx 32\text{--}40 \text{ s}$.
- **PI-based boost-converter** control architecture (current- or voltage-source).

A simple model, calibrated per cell, that is faithful enough to ground a real control design.

- Implement & **online-tune** the boost PI loop on the bench; hydrogen-pressure regulator with a reduced-order observer.
- Control upgrades: **LQ**, time-delay, **MPC**.
- Richer model: oxygen transients, liquid-water accumulation, **gas-supply delays**.
- Replace degraded cells → stack-level closed-loop experiments.

Ref.: [9] Pukrushpan 2003; [7] Kim & Kang 2010; [13] Wang et al. 2021; [6] Kharrat & Haußmann 2025

- [1] A. E. Andrea et al., *A simplified electrical model of small PEM fuel cell*, RE&PQJ, 2006.
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- [6] J. Kharrat, J. Haußmann, *Influence of the gas supply component delay on the dynamic operation of the PEM fuel cell system*, Hydrogen Tech. Conf., 2025.
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- [8] A. K. M. Mohiuddin, N. Basran, A. A. Khan, *Modelling and validation of PEMFC*, IOP Conf. Ser. MSE, 2018.
- [9] J. T. Pukrushpan, *Modeling and Control of Fuel Cell Systems and Fuel Processors*, PhD thesis, Univ. of Michigan, 2003.

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- [11] K. K. T. Thanapalan, J. G. Williams, G. P. Liu, D. Rees, *Modelling of a PEM fuel cell system*, 17th IFAC World Congress, 2008.
- [12] A. Thomya, Y. Khunatorn, *Design of control system of hydrogen and oxygen flow rate for PEMFC using fuzzy logic controller*, Energy Procedia, 2011.
- [13] Y. Wang et al., *Simulation study on PEMFC oxygen starvation based on coupling MPC and PID*, Energy Conversion and Management, 2021.
- [14] A. Z. Weber, J. Newman, *Modeling transport in polymer-electrolyte fuel cells*, Chemical Reviews, 2004.
- [15] G. Zhang, L. Fan, J. Sun, K. Jiao, *A 3D model of PEMFC considering detailed multiphase flow and anisotropic transport properties*, Int. J. Heat and Mass Transfer, 2017.
- [16] Y. Zhu et al., *Modelling and fuel flow control of PEMFC considering over-pressure case*, IEEE ICIEA, 2017.

The complete source code developed during this internship — model, identification scripts, processing pipelines, and the report/presentation sources — is available on GitHub:

https://github.com/KhalilEllahLAGHA/PEMFC_Internship_codes

Thank you for your attention

Questions?

Khalil Ellah Lagha — Université Grenoble Alpes / GIPSA-lab
Supervisor: Prof. Emmanuel Witrant

$$\Delta g_f = \Delta g_f^0 - \bar{R} T_{fc} \ln \frac{p_{H_2} p_{O_2}^{1/2}}{p_{H_2O}}, \quad \Delta g_f = -2FE$$

$$E = -\frac{\Delta g_f^0}{2F} + \frac{\bar{R} T_{fc}}{2F} \ln \frac{p_{H_2} p_{O_2}^{1/2}}{p_{H_2O}}$$

- Standard voltage $E^0 \approx 1.229 \text{ V}$ (liquid water, 25°C).
- Temperature coefficient $\approx -0.85 \times 10^{-3} \text{ V/K}$.
- Nernst correction $\approx 4.31 \times 10^{-5} T_{fc}$.

(Report Appendix A / Table A.1.)

Backup B2 — Full Per-Cell Parameter Table

Cell	RMSE [mV]	ξ_1	ξ_2	ξ_3	ξ_4	B
0 [†]	0.08	-7.783	1.67×10^{-2}	-6.06×10^{-4}	≈ 0	1.08×10^{-2}
1 [†]	0.08	-7.783	1.67×10^{-2}	-6.06×10^{-4}	≈ 0	1.08×10^{-2}
2 [†]	0.08	-7.783	1.67×10^{-2}	-6.06×10^{-4}	≈ 0	1.08×10^{-2}
3	4.66	-1.977	2.83×10^{-2}	1.32×10^{-3}	-3.17×10^{-4}	≈ 0
4	2.18	0.386	1.43×10^{-2}	8.11×10^{-4}	-1.24×10^{-4}	≈ 0
5	4.23	0.682	3.00×10^{-2}	1.93×10^{-3}	-2.98×10^{-4}	≈ 0
6	4.51	1.775	4.62×10^{-3}	4.87×10^{-4}	-2.73×10^{-4}	≈ 0
7	5.62	1.796	2.54×10^{-2}	1.82×10^{-3}	-1.99×10^{-4}	≈ 0
8	1.96	-4.946	2.81×10^{-2}	6.92×10^{-4}	-1.34×10^{-4}	≈ 0
9	11.25	-5.400	2.01×10^{-2}	9.31×10^{-5}	-1.06×10^{-4}	3.21×10^{-7}

[†] Degraded cell (voltage collapse under load); fit not physically meaningful.

$$\begin{aligned} \frac{\partial v_{fc}}{\partial P_{H_2}} &= \frac{4.31 \times 10^{-5} T}{P_{H_2,0}}, & m_w \rightarrow \phi \rightarrow \lambda_m \rightarrow \sigma_m \rightarrow v_{ohm} \\ \frac{\partial v_{fc}}{\partial i_{FC}} &= -\left(\frac{\xi_4 T}{i_0} + \rho_{m,0} \frac{t_m}{A_{fc}} + c + \frac{B}{i_{\max} - i_0} \right), & \frac{\partial v_{fc}}{\partial W_{H_2,in}} = 0 \\ C &= \begin{bmatrix} \frac{4.31 \times 10^{-5} T}{P_{H_2,0}} & \kappa_0 \lambda'(\phi_{an,0}) \frac{R_v T}{2V_{an} p_{sat}} & \kappa_0 \lambda'(\phi_{ca,0}) \frac{R_v T}{2V_{ca} p_{sat}} \end{bmatrix} \end{aligned}$$

With λ_m held constant, the water entries vanish: $C_{red} = [4.31 \times 10^{-5} T / P_{H_2,0} \quad 0 \quad 0]$.

Averaging the switch-ON / switch-OFF topologies over one PWM period, with state $x = [i_L \ V_{OUT}]^T$:

$$\dot{x} = \begin{bmatrix} 0 & \frac{T-1}{L} \\ \frac{1-T}{C} & -\frac{1}{RC} \end{bmatrix} x + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} u, \quad y = \begin{bmatrix} 0 & 1 \end{bmatrix} x$$

$$V_{out} = \frac{1}{1-T} V_{in}, \quad T = \text{duty cycle} \in [0, 1)$$

Static voltage gain: $T = 0.5 \Rightarrow V_{out}/V_{in} = 2$; $T = 0.6 \Rightarrow 2.5$.